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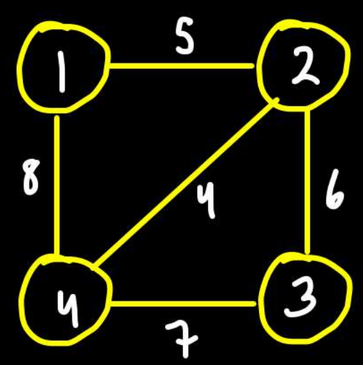
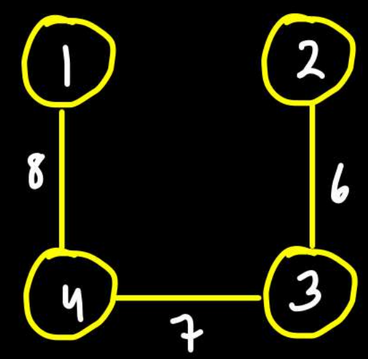
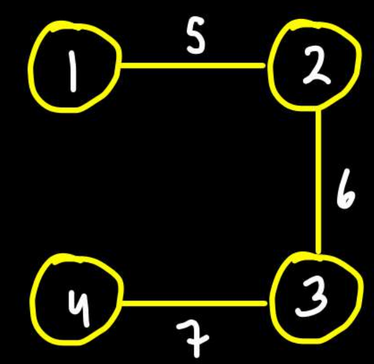
nodes = 4
edges = 5

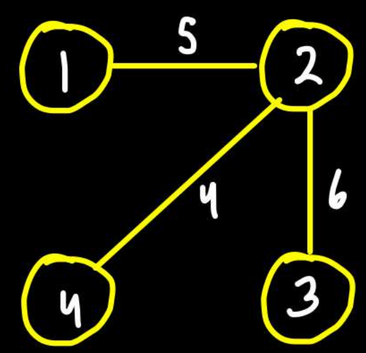


Spanning Tree → if 'n' nodes then 'n-1' edges





minimum spanning tree



.. & so on

edges sum = 15

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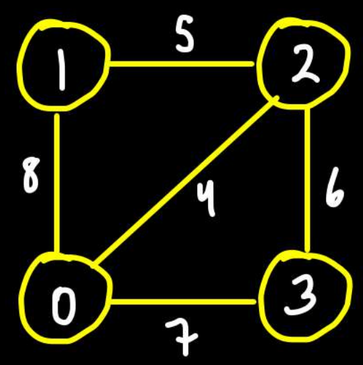
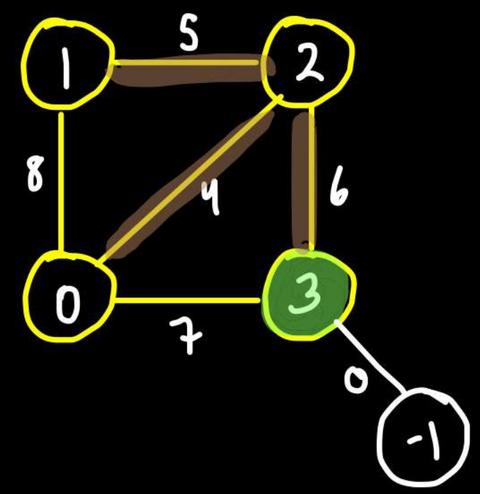
MST sum = ?

Prim's Algorithm → Dijkstra's → (PQ)



(0, ~~3~~, ~~7~~) (~~1~~, ~~0~~, 8)

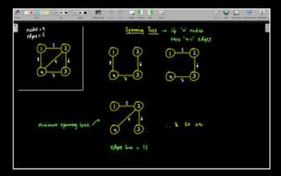


pq <Node, parent, wt>

minheap

	0	1	2	3
vis =	T	T	T	T

Sum = ~~0~~ ~~4~~ ~~6~~ ~~16~~ 15



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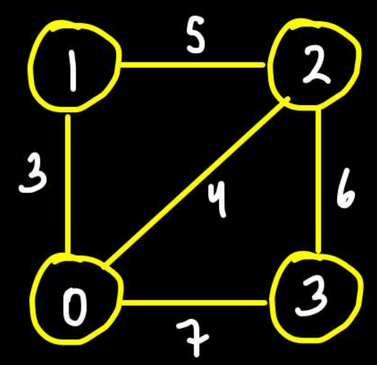
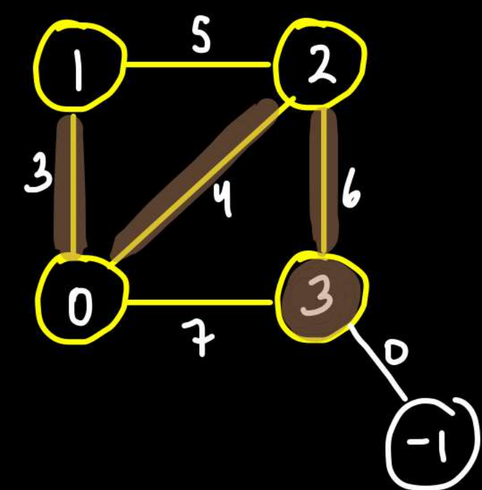
MST sum = ?

Prim's Algorithm → Dijkstra's → (PQ)



0 : (1,3), (2,4), (3,7)
1 : (0,3), (2,5)
2 : . .



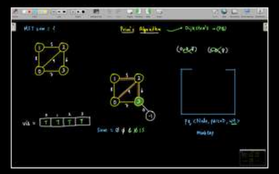
vis =

0	1	2	3
T	T	T	T

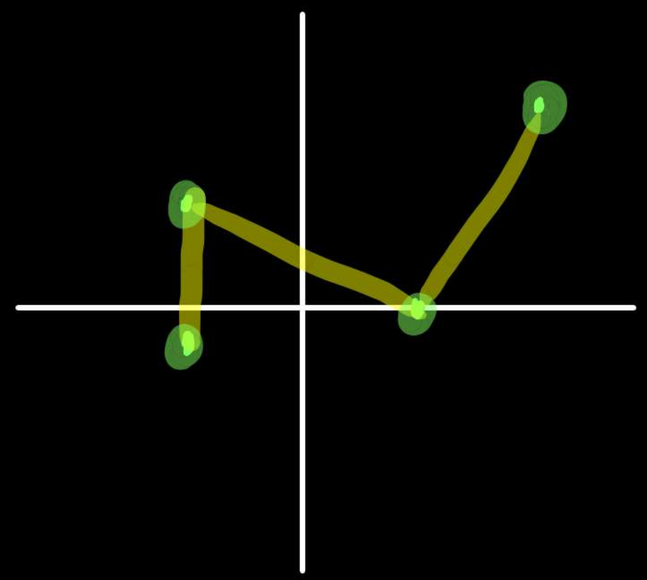
Sum = 0 + 0 + 6 + 4 + 3

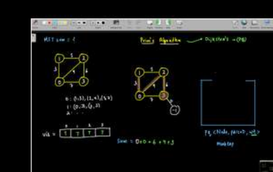
pq <Node, parent, wt>

minheap



Q, Minimum cost to connect all houses in a city.





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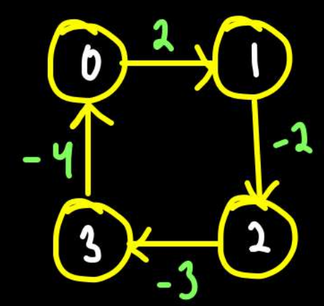
Bellman Ford Algorithm → Brute Force

What & Why? → Shortest distances from src to everyone else, graph with negetive wt. with negetive cycle

Works in DG / UG

→ again & again we search for min distances ↓ relaxation

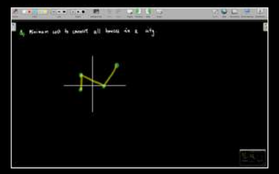


0 to 1 ka shortest dist → 2

↓

0 - 1 - 2 - 3 - 0 - 1

$\cancel{2} - \cancel{2} - 3 - 4 + 2 = -5$



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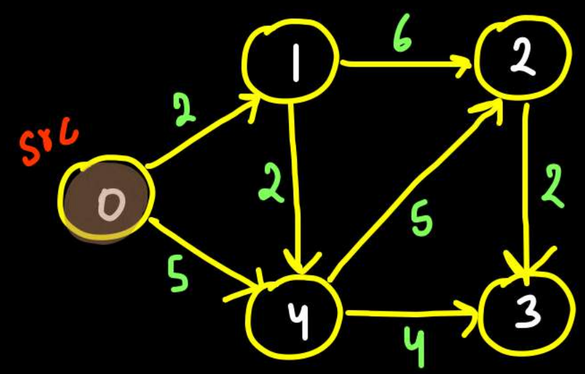
n-1 times relaxation

Bellman Ford Algorithm

→ Brute Force



	0	1	2	3	4
dist	0	2	8	~~8~~ 8	~~8~~ 4

[u,v,w]
if (dist[u] + wt < dist[v])
update dist[v]

edge list

[4, 3, 4]
[1, 4, 2]
[1, 2, 6]
[0, 1, 2]
[4, 2, 5]
[0, 4, 5]
[2, 3, 2]

src $\xrightarrow{4}$ 4 $\xrightarrow{4}$ 3 (4+4 < 9) ✓
src $\xrightarrow{2}$ 1 $\xrightarrow{2}$ 4 ✗
src $\xrightarrow{2}$ 1 $\xrightarrow{6}$ 2 ✗
src $\xrightarrow{0}$ 0 $\xrightarrow{2}$ 1 ✗
src $\xrightarrow{4}$ 4 $\xrightarrow{5}$ 2 ✗
src $\xrightarrow{0}$ 0 $\xrightarrow{5}$ 4 ✗
src $\xrightarrow{8}$ 2 $\xrightarrow{2}$ 3 ✗

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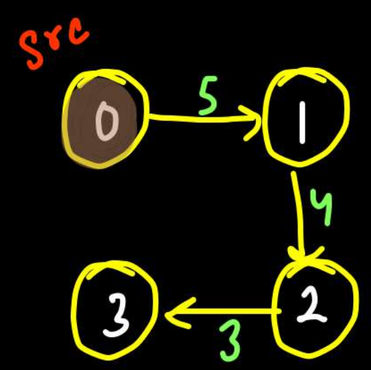
Why 'n-1' times (Worst case scenario)



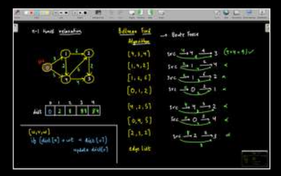
	0	1	2	3
dist	0	5	9	12

nodes = 4 = n

[2, 3, 3] src $\xrightarrow{1}$ 2 $\xrightarrow{3}$ 3 (12 < ∞) ✓

[1, 2, 4] src → 1 $\xrightarrow{4}$ 2

[0, 1, 5] src → 0 $\xrightarrow{5}$ 1



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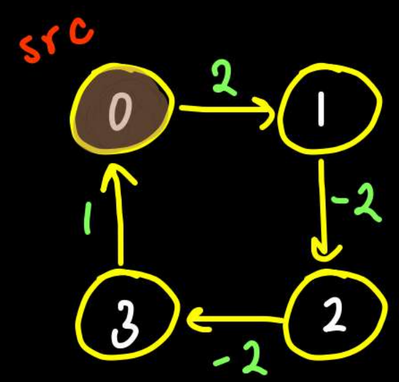
Negetive wt Example

cycle Sum = -1



[0, 1, 2] src $\xrightarrow{-1}$ 0 $\xrightarrow{2}$ 1, 0 $\xrightarrow{2}$ 1 (1 < 2) ✓ → Gadbad

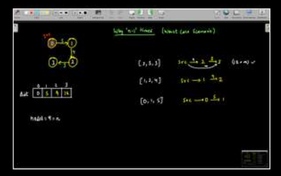
[3, 0, 1] src → 3 $\xrightarrow{1}$ 0

[2, 3, -2] src → 2 $\xrightarrow{-2}$ 3

[1, 2, -2] src → 1 $\xrightarrow{-2}$ 2

n = 4

dist

0	1	2	3
~~0~~	~~∞~~	~~∞~~	~~∞~~
-1	~~2~~ 1	0	-2

n^{th} time iterate relaxation